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NON-EXPLOSIVITY OF ENDOTACTIC STOCHASTIC REACTION SYSTEMS

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Abstract. Reaction networks have been widely used as generic models in diverse areas of applied science, such as biology, chemistry, ecology, epidemiology, and computer science. Reaction networks incorporating noisy effect are modelled as continuous time Markov chains (CTMC), and are called *stochastic reaction systems*. *Non-explosivity* is a concept that characterizes regularity of CTMCs. In this paper, we study non-explosivity of stochastic reaction systems, in the sense of their underlying CTMCs. By constructing a simple linear Lyapunov function, we obtain non-explosivity for a class of *endotactic* stochastic reaction systems containing second-order endotactic stochastic mass-action systems as a subset. As a consequence, we prove that every bimolecular weakly reversible stochastic mass-action system is non-explosive. We apply our results to diverse models in biochemistry, epidemiology, ecology, and synthetic biology in the literature.

1. Introduction

Reaction networks are widely used in modelling diverse science phenomena [17]. The original motivation seems to come from biochemistry, and hence the mathematical theory of reaction networks is conventionally referred to as Chemical Reaction Network Theory (CRNT) [17]. When the molecule counts of species are low, reaction networks are modelled as stochastic systems in terms of continuous time Markov chains (CTMCs) [8]. In contrast, when species are abundant [8], reaction networks are modelled as deterministic systems in terms of ordinary differential equations (ODEs), as mean field approximations of stochastic systems as the *volume* of the reactor tends to infinity.

In CRNT, a particular focus is given to *bimolecular* reaction networks where *molecularity* (see Section 2 for the definition) of each reaction is no more than two [48, 2]. This is because, in practice in e.g. biochemistry, it is less likely for more than two molecules to bind together by collision to form a complex as the reaction may need much more activation energy to proceed over a chemical potential well [12, Chapter 14]. For that reason, for any reaction, a product with more than two molecules may be assumed to first dissociate spontaneously thanks to a large amount of internal energy before stabilizing by collision. Due to this consideration, reactions of bigger molecularities are usually interpreted as a sequence of bimolecular reactions that proceed in multiple steps [19], in analogy to that natural numbers bigger than one can be represented as binary numbers. This analogy might explain why the term "bimolecular reaction" may be used interchangeably with *"binary reaction"* in the literature of CRNT [6].

Positive recurrence, among many fundamental properties of CTMCs such as *structure decomposition* [25, 56], *recurrence and transience* [6, 3, 57], is desirable for stochastic reaction systems [8, 17], which means the underlying CTMC starting from any state in a closed communicating class, will recur with probability one in finite expected time. A positive recurrence conjecture was proposed by Anderson and Kim in 2018 [6] that *every weakly reversible stochastic mass-action system is positive recurrent* on each closed communicating class. Consistent progress has been made ever since [6, 1, 2, 3, 57, 53, 7, 37, 34, 15]. A classical tool to attack this problem relies on the so-called Foster-Lyapunov criterion [42, 43, 41].

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Primarily based on this criterion, continued efforts since [6] have been made towards this conjecture [6, 2, 57, 53, 7]. Likely in the light of the aforementioned practical considerations from biochemistry, this conjecture was first proved with diverse novel techniques such as tier-sequence, for *bimolecular systems* under various additional assumptions [6, 2, 3]. Moreover, since *complex-balancing* [32] is also a pleasant property in biochemistry, it is interesting to know if a complex-balanced stochastic reaction system is also positive recurrent [4]. It was proved that every complex-balanced stochastic mass-action system is non-explosive [4, Theorem 3]; in the light of the Poisson stationary distribution for every complex-balanced mass-action stochastic system [5], these systems are positive recurrent [45, Theorem 3.5.3]. This further implies that every weakly reversible *deficiency zero* stochastic mass-action systems (for the definition of deficiency, see the footnote in Example 3.15) are also positive recurrent on the grounds that they are complex-balanced [32]. As a consequence, the conjecture was also confirmed for *unimolecular* systems (i.e., every non-zero reactant is a copy of one species), since every unimolecular weakly reversible reaction network is of deficiency zero [5].

Later, this conjecture was proved in one-dimension in [53], the arguably simplest scenario, based on a handy criterion for positive recurrence of one-dimensional CTMCs of polynomial transition rates recently established in [57]. A more exciting progress on this conjecture was recently announced by Daniele Cappelletti and by Andrea Aggazi in their respective talks: With their collaborators David Anderson and Jonathan Mattingly, they managed to prove this conjecture in two-dimensions by introducing new techniques including a *relaxed Foster-Lyapunov drift condition* as well as a *very delicate construction of Lyapunov functions*.

Nevertheless, in spite of all these efforts, the conjecture in the bimolecular case has yet to be closed. In this paper, even though we are still unable to prove every bimolecular weakly reversible stochastic mass-action system is positive recurrent, we indeed show that every such system is non-explosive (Corollary 3.3(iv)). This result helps reduce this problem of proving positive recurrence (for bimolecular weakly reversible stochastic mass-action systems) to finding a stationary distribution for these systems (Corollary 3.8). Hence rather than constructing Lyapunov functions, one may count on alternative techniques to prove the existence of stationary distributions [5, 9, 31, 35, 29, 30, 27, 54] to show positive recurrence for bimolecular systems, as illustrated by Example 3.15.

A wider class of reaction networks than weakly reversible reaction networks are *endotactic reaction networks*. Endotactic reaction networks were originally introduced in [13], to study *permanence* of reaction networks in the deterministic regime. Despite endotacticity and strong endotacticity (see Section 2 for their definitions) have been shown to be sufficient for permanence of deterministic mass-action reaction systems [47, 13, 23, 55] in two dimensions and in arbitrary finite dimensions, respectively, they are in general *insufficient* for non-explosivity or recurrence in the stochastic regime, as evidenced by counterexamples in terms of two-species stochastic mass-action systems [1, 3]. Hence it is interesting to know if other factors (e.g., *order*, the maximal molecularity of all reactions for stochastic mass-action systems) of reaction networks might jointly play a decisive role with endotacticity in determining non-explosivity of endotactic stochastic mass-action systems. In this paper, using a simple linear Lyapunov function, we show that a class of stochastic reaction systems including *second order* endotactic stochastic mass-action systems as a subset, are non-explosive (Theorem 3.1). Despite linear functions turn out to be Lyapunov functions for non-explosivity of second order weakly reversible stochastic mass-action systems, in general they fail to work for *termolecular* weakly reversible stochastic mass-action systems (i.e., molecularity of each reaction is no more than three) evidenced by Example 3.14. This implies that even if every weakly reversible stochastic mass-action system were indeed non-explosive [6], it could be challenging to construct a *simple* Lyapunov function to prove non-explosivity of weakly reversible stochastic mass-action systems of a higher order.

2. Preparations

Notation. Let $\mathbb{R}$ be the set of real numbers, $\mathbb{R}_+$ the set of non-negative real numbers, and $\mathbb{R}_{++}$ the set of positive real numbers. Let $\mathbb{N}$ and $\mathbb{N}_0$ denote the set of positive integers and non-negative integers, respectively. Let $\{e_i\}_{i=1}^d$ be the standard basis of $\mathbb{R}^d$ and $\mathbf{1}_d = \sum_{i=1}^d e_i$. We simply write $\mathbf{1}$ short for $\mathbf{1}_d$ when d is deemphasized and clear from the context. We denote $\|x\|_1$ the ℓ_1-norm of a vector $x \in \mathbb{R}^d$.

Reaction networks. A reaction network is a triple of sets: a set of species $\mathcal{S}$, a set of complexes $\mathcal{C}$, and a set of reactions $\mathcal{R}$. Species are barely symbols, which are abstracted from chemical species. Each complex is a linear combination of species with non-negative integer coefficients (called *stoichiometric coefficients*), and $\mathcal{R}$ is a relation on $\mathcal{C}$ with each reaction denoted $y \longrightarrow y'$, where the complexes y (called the *reactant*) and y' (the *product*) are *distinct*, and $y'-y$ the *reaction vector*. Hence a reaction network of d species can be represented as a simple directed graph $(\mathcal{C}, \mathcal{R})$ without *isolated* vertices (vertices of zero in-degree and zero out-degree), called *reaction graph*, which can be embedded in $\mathbb{N}_0^d$. Every complex is identified with a vector of its stoichiometric coefficients in $\mathbb{N}_0^d$, e.g., $S_1 + 2S_2$ is identified with $e_1 + 2e_2$, and thus both notations will be interchangeably used to denote a complex throughout. A reaction network is *weakly reversible* if its reaction graph is a union of pairwise disjoint strongly connected components. Let $\mathcal{C}_+$ be the set of reactants, which usually determines the *kinetics* of a reaction network of types of interest including the *mass-action kinetics* defined below. Throughout this paper, we assume by convention that *every species appears in at least one complex*. Hence the three sets constituting a reaction network are all known from the set of reactions. For this reason, we hereafter will simply use $\mathcal{R}$ to represent a reaction network consisting of reactions in $\mathcal{R}$. The linear span of the set of reaction vectors is called the *stoichiometric subspace* of the reaction network. The dimension of the stoichiometric subspace of a reaction network is also simply called the *dimension of the reaction network*. A positive vector in $\mathbb{R}_{++}^d$ is a *conservation law* [44] of a reaction network if it is in the orthogonal complement of the stoichiometric subspace of the reaction network.

For every reaction $y \to y' \in \mathcal{R}$, the sum of stoichiometric coefficients of the reactant (i.e., ℓ_1-norm of the reactant $\|y\|_1 := \sum_{j=1}^d y_j$) is called the *molecularity* of the reaction. If the molecularity of every reaction is no more than one/two/three, then the reaction network is called *unimolecular/bimolecular/termolecular*, respectively. A *stochastic kinetics* of a reaction network $\mathcal{R}$ is a collection of *propensity functions* $\Lambda = \{\lambda_{y\to y'} \colon \mathbb{N}_0^d \to \mathbb{R}_+ \colon y \to y' \in \mathcal{R}\}$, each of which corresponds to one reaction in the network, characterizing how likely that reaction may fire. A reaction $y \longrightarrow y' \in \mathcal{R}$ is *active* in a state x if $\lambda_{y\to y'}(x) > 0$, and is *inactive* otherwise. Furthermore, a reaction is *active/inactive* in a set if it is active/inactive in every state in the set. A reaction network endowed with a stochastic kinetics, denoted as a pair $(\mathcal{R}, \Lambda)$, is called a *stochastic reaction system* (SRS). For a majority of kinetics of interest in CRNT, propensity function of each reaction in a reaction network only depends on the reactant of the reaction. One particular such stochastic kinetics is the *stochastic mass-action kinetics* defined as follows. Given the reaction network $\mathcal{R}$, for every $y \to y' \in \mathcal{R}$,

$$\lambda_{y\to y'}(x) = \kappa_{y\to y'} x^{\underline{y}}, \quad x \in \mathbb{N}_0^d, \tag{2.1}$$

where $\kappa_{y\to y'}$ is the *reaction rate constant*, $x^{\underline{y}} = \prod_{i=1}^d x_i^{\underline{y_i}}$ for $x \geq y$ (coordinate-wise); $x^{\underline{y}} = 0$ otherwise, and $a^{\underline{b}} = \frac{a!}{(a-b)!}$ is the falling factorial for $a, b \in \mathbb{N}_0$ and $a \geq b$. A stochastic reaction system of stochastic mass-action kinetics is called a *stochastic mass-action system* (SMAS). Hence saving the symbol Λ of kinetics, a stochastic mass-action system can be identified with a *weighted* directed graph $\mathcal{R}$ embedded in $\mathbb{N}_0^d$, where the positive weight assigned to each edge in the graph is the reaction rate constant associated with that reaction in the reaction network.

Under stochastic mass-action kinetics, the sum of exponents in the propensity function of a reaction is referred to as the *order* of the reaction, and hence it coincides with the molecularity of that reaction. The maximal order of all reactions of a stochastic mass-action system $\mathcal{R}$ is called the *order* of $\mathcal{R}$. For instance, second order stochastic mass-action systems are bimolecular, and *termolecular* stochastic mass-action systems are of third order. Despite there is no distinction between order and molecularity of a reaction for a stochastic mass-action system, in general they may *not* coincide, since *molecularity applies to the molecular mechanism of an elementary step of a reaction while in contrast, order refers to the empirical rate law of a reaction* [40].

Modelling of an SRS. An SRS $(\mathcal{R}, \Lambda)$ can be modelled in terms of a stochastic differential equation [8]:

$$X_t = X_0 + \sum_{y \to y' \in \mathcal{R}} Y_{y \to y'}\big(\int_0^t \lambda_{y \to y'}(X_s) ds\big) \tag{2.2}$$

where X_t is the vector of molecule counts of the d species at time t, $Y_{y \to y'}$ are independent unit Poisson processes associated with each reaction in the SRS, $X_0 \in \mathbb{N}_0^d$ is the deterministic initial vector of species counts, and the propensity functions $\lambda_{y \to y'}$ are defined in (2.1). Note that the *independence* of these Poisson processes is a consequence of the assumption that *reactions cannot fire simultaneously* (with a positive probability). Hence X_t is a CTMC on a state space as a subset of $\mathbb{N}_0^d$ [14] with the extended infinitesimal generator

$$\mathcal{L}_{\mathcal{R}} f(x) = \sum_{y \to y' \in \mathcal{R}} \lambda_{y \to y'}(x)(f(x + y' - y) - f(x)),$$

for every real-valued function f defined on $\mathbb{N}_0^d$.

Endotacticity. Next, we recall the definitions of endotactic and strongly endotactic reaction networks. Both reaction networks were originally proposed [13, 23] to address dissipativity of the corresponding deterministic reaction systems.

Given $u \in \mathbb{R}^d$, for any $y, z \in \mathbb{R}^d$, we write $y >_u z$ or $z <_u y$ if $(y - z)^T \cdot u > 0$. We similarly define $y =_u z$, $y \geq_u z$, and $z \leq_u y$. In this way, u induces a total order on $\mathbb{R}^d$. Two vectors $y, z \in \mathbb{R}$ are *u-equal* if $y =_u z$. Given any subset $A \subseteq \mathbb{R}$, an element $y \in A$ is *u-maximal* in A if $y \geq_u z$ for all $z \in A$. Let $\mathcal{R}_u$ be the (possibly empty) set of reactions whose reaction vectors are *not* orthogonal to u and $\mathcal{C}_{u,+}$ be the (possibly empty) set of sources of reactions in $\mathcal{R}_u$.

Definition 2.1. Given a reaction network $\mathcal{R}$, let $u \in \mathbb{R}^d$. Any reaction $y \longrightarrow y' \in \mathcal{R}$ with a u-maximal source in $\mathcal{C}_{u,+}$ satisfying $y <_u y'$ is called a *u-endotacticity violating reaction* of $\mathcal{R}$, or simply called *endotacticity violating reaction* of $\mathcal{R}$. We say $\mathcal{R}$ is *u-endotactic* if it has no u-endotacticity violating reaction. A u-endotactic reaction network $\mathcal{R}$ is *strongly u-endotactic* if there exists a reaction $y \to y' \in \mathcal{R}$ such that $y >_u y'$ and y is u-maximal in $\mathcal{C}_+$. Furthermore,

(i) $\mathcal{R}$ is *lower-endotactic* (*strongly lower-endotactic*, respectively) if $\mathcal{R}$ is u-endotactic (strongly u-endotactic, respectively) for every $u \in \mathbb{R}_+^d$;

(ii) $\mathcal{R}$ is *endotactic* (*strongly endotactic*, respectively) if $\mathcal{R}$ is u-endotactic (strongly u-endotactic, respectively) for every $u \in \mathbb{R}^d$.

In other words, *$\mathcal{R}$ is endotactic if and only if $\mathcal{R}$ has no endotacticity-violating reaction.*

For every $u \in \mathbb{R}_{++}^d$, let $\mathcal{R}_{u,+} := \{y \longrightarrow y' \in \mathcal{R} \colon y' >_u y\}$ be the (possibly empty) subset of reactions as bona fide vectors increasing along u-direction. Note that $\mathcal{R}_{u,+}$ may be non-empty for an endotactic reaction network, consisting of reactions of *non* u-maximal reactants in $\mathcal{C}_{u,+}$.

Only based on this definition, it may turn out non-trivial to justify that a reaction network is endotactic, since one literally needs to show u-endotacticity for infinitely many vectors u. However, for either 2-dimensional reaction networks [13, Proposition 4.1] or first order reaction networks [55, Theorem 5.2], to conclude endotacticity of a reaction network, it suffices to check u-endotacticity for only finitely many vectors u. Reaction networks satisfying other more readily verifiable conditions purely on the reaction graph are known to be endotactic [55, Theorem 4.4]; for instance, *every weakly reversible reaction network is endotactic* [13, Lemma 4.5].

Since the main results of this paper impose conditions on u-endotacticity for a particular vector u, to save some extra notations, we will omit stating the known criteria for endotacticity of reaction networks particularly for 2-dimensional or first order reaction networks. We refer the interested reader to [13, 23, 55] for more details.

Example 2.2. Consider the following Brusselator reaction network [16]:

$$\mathcal{R}\colon 0 \rightleftharpoons S_1 \longrightarrow S_2 \quad 2S_1 + S_2 \longrightarrow 3S_1$$

(i) Let $u = (1,1)$. Then $\mathcal{R}_u = \{0 \rightleftharpoons S_1\}$, and hence $\mathcal{C}_{u,+} = \{0, e_1\}$. Since $e_1 >_u 0$, there exists no u-endotacticity violating reaction. Hence $\mathcal{R}$ is u-endotactic.

(ii) Let $v = (2,1)$. Then $\mathcal{R}_v = \mathcal{R}$ since no reaction vector is orthogonal to v. Moreover, $\mathcal{C}_{v,+} = \{0, e_1, 2e_1 + e_2\}$, and $2e_1 + e_2$ is the unique v-maximal source in $\mathcal{C}_{v,+}$. Observe that $2S_1 + S_2 \longrightarrow 3S_1$ is a v-endotacticity violating reaction since $2e_1 + e_2 <_v 3e_1$, which yields that $\mathcal{R}$ is not (v-)endotactic. In addition, it is straightforward to show that

$$\mathcal{R}_{v,+} = \{0 \longrightarrow S_1 \quad 2S_1 + S_2 \longrightarrow 3S_1\}$$

3. Non-explosivity

Let $(\mathcal{R}, \Lambda)$ be an SRS. Let X_t be the solution to (2.2) subject to the initial condition $X_0 \in \mathbb{N}_0^d$. We say X_t *explodes* at X_0 if X_t jumps infinitely many times within a finite time interval with a positive probability. We say the SRS $(\mathcal{R}, \Lambda)$ is *explosive* if X_t explodes at some $X_0 \in \mathbb{N}_0^d$, and is *non-explosive* otherwise. Non-explosivity is a property which can be determined by the infinitesimal generator, or equivalently, the Q-matrix of the underlying CTMC X_t [45, Corollary 2.7.3]. Hence the set Λ of propensity functions uniquely determines the non-explosivity of $(\mathcal{R}, \Lambda)$. Similarly, an SRS is *transient/recurrent/positive recurrent* in a communicating class Γ of $\mathbb{N}_0^d$ if every state in Γ is transient/recurrent/positive recurrent for the underlying CTMC, respectively. Note that these three properties are *class* properties of CTMCs [45]. In particular, an SRS is *transient/recurrent/positive recurrent* if it is transient/recurrent/positive recurrent in every communicating class, respectively.

Next, we identify a class of non-explosive SRS in terms of a simple condition. Roughly speaking, an SRS is non-explosive if there exists a positive direction such that all reactions that increase along this direction have non-strictly sub-linear propensity functions.

We hereafter deemphasize the number d of species of a stochastic reaction system, and accordingly, we write $\mathbf{1}$ for $\mathbf{1}_d$ for simplicity.

Theorem 3.1. *Let $(\mathcal{R}, \Lambda)$ be an SRS with $\Lambda = \{\lambda_{y \to y'} \colon y \longrightarrow y' \in \mathcal{R}\}$. If there exists a positive vector $u \in \mathbb{R}_{++}^d$ and a constant $C > 0$ such that*

$$\sup_{y \to y' \in \mathcal{R}_{u,+}} \sup_{x \in \mathbb{N}^d \setminus \{0\}} \frac{\lambda_{y \to y'}(x)}{\|x\|_1} \leq C, \tag{3.1}$$

then $(\mathcal{R}, \Lambda)$ is non-explosive. In particular, $(\mathcal{R}, \Lambda)$ is non-explosive if $\mathcal{R}_{u,+} = \emptyset$ for some $u \in \mathbb{R}_{++}^d$.

Proof. Let $V(x) = x^T \cdot u$. It follows from (3.1) that

$$\begin{aligned}\mathcal{L}V(x) &= \sum_{y\to y'\in\mathcal{R}} \lambda_{y\to y'}(x)\Big((x+y'-y)^T\cdot u) - x^T\cdot u\Big)\\ &\le \sum_{y\to y'\in\mathcal{R}} \lambda_{y\to y'}(x)(y'-y)^T\cdot u\\ &\le \sum_{y\to y'\in\mathcal{R}_{u,+}} \lambda_{y\to y'}(x)(y'-y)^T\cdot u \le \widetilde{C}V(x)\end{aligned}$$

for some constant $\widetilde{C}\in\mathbb{R}$. Hence by Proposition A.1, $(\mathcal{R},\Lambda)$ is non-explosive.

Note that $\sup_{y\to y'\in\mathcal{R}_{u,+}}\sup_{x\in\mathbb{N}^d\setminus\{0\}}\frac{\lambda_{y\to y'}(x)}{\|x\|_1} = 0$ if $\mathcal{R}_{u,+}=\emptyset$. This immediately yields that $(\mathcal{R},\Lambda)$ is non-explosive. □

Corollary 3.2. *Every first order stochastic mass-action system is non-explosive.*

Proof. Let $\mathcal{R}$ be a first order stochastic mass-action system. Then by finiteness of $\mathcal{R}$, there exists $C>0$ such that

$$\sup_{x\in\mathbb{N}^d\setminus\{0\}}\frac{\lambda_{y\to y'}(x)}{\|x\|_1}\le C,\quad \forall y\to y'\in\mathcal{R},$$

which yields (3.1). Hence the non-explosivity of $\mathcal{R}$ follows from Theorem 3.1. □

Corollary 3.3. *Every second order* **1***-endotactic stochastic mass-action system is non-explosive. In particular,*

(i) *every second order lower-endotactic stochastic mass-action system is non-explosive.*
(ii) *every second order endotactic stochastic mass-action system is non-explosive.*
(iii) *every second order strongly lower-endotactic stochastic mass-action system is non-explosive.*
(iv) *every second order strongly endotactic stochastic mass-action system is non-explosive.*
(v) *every second order weakly reversible stochastic mass-action system is non-explosive.*

Proof. Let $\mathcal{R}$ be a second order **1**-endotactic stochastic mass-action system. By **1**-endotacticity of $\mathcal{R}$, for every second order reaction $y\longrightarrow y'\in\mathcal{R}$, we have

$$(y'-y)^T\cdot\mathbf{1}\le 0$$

Hence (3.1) holds with $u=\mathbf{1}$ and some positive constant C. By Theorem 3.1, $\mathcal{R}$ is non-explosive. The rest conclusions in (i)-(v) follow since these networks are all **1**-endotactic [13, 23]. □

Remark 3.4. In [3, Example 3.2], an explosive strongly endotactic reaction network of order 10 is constructed. It remains open *what is the minimal order for an endotactic stochastic mass-action system to be explosive.*

Note that **1**-endotacticity is not necessary while crucial for non-explosivity.

Example 3.5. Consider the stochastic mass-action system

$$\mathcal{R}\colon S_1 \underset{\kappa_{-2}}{\overset{\kappa_1}{\rightleftharpoons}} 2S_1 \xrightarrow{\kappa_2} 3S_1$$

It is easy to verify that $\mathcal{R}$ is *not* 1-endotactic. Applying [57, Theorem 1] (see also [53, Theorem 4.4(i)]), a necessary and sufficient condition for non-explosivity of one-dimensional CTMCs of polynomial transition rates, we know that $\mathcal{R}$ is non-explosive if and only if $\kappa_2\le\kappa_{-2}$.

For higher (than 2) order stochastic mass-action systems, even *strong* **1**-endotacticity becomes insufficient for non-explosivity.

Example 3.6. Consider the following termolecular reaction network

$$\mathcal{R}\colon 0 \underset{\kappa_{-1}}{\overset{\kappa_0}{\rightleftharpoons}} S_1 \quad 2S_1 \xrightarrow{\kappa_1} 3S_1 \quad 3S_2 \xrightarrow{\kappa_2} 2S_2$$

It is readily verified that $\mathcal{R}$ is strongly $(1,1)$-endotactic. Nevertheless, $\Gamma_i = \mathbb{N} \setminus \{1\} \times \{i-1\}$ for $i = 1,2,3$ are the only three closed communicating classes in the ambient space $\mathbb{N}_0^2$, and the reaction $3S_2 \xrightarrow{\kappa_2} 2S_2$ is inactive in $\Gamma = \cup_{i=1}^3 \Gamma_i$, and hence the infinitesimal generator of $\mathcal{R}$ coincides with that of

$$\mathcal{R}'\colon 0 \underset{\kappa_{-1}}{\overset{\kappa_0}{\rightleftharpoons}} S_1 \quad 2S_1 \xrightarrow{\kappa_1} 3S_1$$

in Γ, which, as a one-dimensional stochastic mass-action system, is explosive [53, Theorem 4.4(i)]. Hence the stochastic mass-action system $\mathcal{R}$ is also explosive.

Corollary 3.7. *Let $\mathcal{R}$ be a second order* **1***-endotactic stochastic mass-action system. Then $\mathcal{R}$ has at most one stationary distribution in each closed communicating class.*

Proof. Assume w.l.o.g. that there exists a closed communicating class Γ and $\mathcal{R}$ has a stationary distribution in Γ. Otherwise the conclusion holds trivially. Then X_t restricted to Γ is irreducible. It follows from Corollary 3.3 that X_t is non-explosive. By [45, Theorem 3.5.3], a non-explosive irreducible CTMC with a stationary distribution is positive recurrent, which implies the uniqueness of the stationary distribution in Γ. □

Corollary 3.8. *Every bimolecular weakly reversible stochastic mass-action system is positive recurrent if and only if there exists a stationary distribution in every communicating class.*

Proof. Note that weakly reversible reaction networks of d species are *essential* in the sense that every communicating class in the ambient state space $\mathbb{N}^d$ is closed [49, Theorem 4.6] (see also [56]). Then the conclusion follows from Corollary 3.7, again in the light of that every weakly reversible reaction network is **1**-endotactic [13, Proposition 4.1]. □

Note that **1***-endotacticity is insufficient for the existence of a closed communicating class.*

Example 3.9. Let

$$\mathcal{R}\colon S_1 + S_2 \longrightarrow 0 \longrightarrow S_2$$

It is readily verified that $\mathcal{R}$ is $(1,1)$-endotactic (indeed, it is lower endotactic). Moreover, the two reactions together facilitate first exhausting species S_1 and then increasing the molecule counts of S_2 indefinitely: Every non-zero state in $\mathbb{N}_0^2$ strictly below the diagonal $\{(n,n)\}_{n\in\mathbb{N}_0}$ leads to a state on the diagonal by possibly repeatedly applying the reaction $0 \longrightarrow S_2$; every state either on or strictly above the diagonal leads to a state on the boundary $\{0\} \times \mathbb{N}_0$ by possibly repeatedly applying the reaction $S_1 + S_2 \longrightarrow 0$; and every state on the boundary $\{0\} \times \mathbb{N}_0$ leads to all states above it by the reaction $0 \longrightarrow S_2$. Hence every state is in an open communicating class and $\mathcal{R}$ is transient [45, Theorem 1.5.5 and Theorem 3.4.1].

In addition, **1**-endotacticity is also insufficient for recurrence (of a closed communicating class, in contrast to the above example) for second order stochastic mass-action systems.

Example 3.10. Consider

$$\mathcal{R}\colon 0 \underset{\kappa_2}{\overset{\kappa_1}{\rightleftharpoons}} S_2 \xrightarrow{\kappa_3} 2S_2 \xleftarrow{\kappa_4} S_1 + S_2 \xrightarrow{\kappa_5} S_2$$

It is straightforward to verify that $\mathcal{R}$ is $(1,1)$-endotactic; moreover $\Gamma = \{0\} \times \mathbb{N}_0$ is the unique closed communicating class, and $\{n\} \times \mathbb{N}_0$ is an open communicating class for every $n \in \mathbb{N}$. Hence restricted to Γ, the infinitesimal generator of $\mathcal{R}$ coincides with that of the sub reaction network of $\mathcal{R}$

$$\mathcal{R}'\colon 0 \underset{\kappa_2}{\overset{\kappa_1}{\rightleftharpoons}} S_2 \xrightarrow{\kappa_3} 2S_2,$$

which is transient if and only if $\kappa_3 > \kappa_2$ or $\kappa_1 > \kappa_2 = \kappa_3$ [57, Theorem 3(i)]. This implies that $\mathcal{R}$ is transient in when $\kappa_3 > \kappa_2$ or or $\kappa_1 > \kappa_2 = \kappa_3$.

Remark 3.11. In [1], a 7th order transient strongly endotactic stochastic mass-action system was constructed. The author is unaware of *what is the minimal order for an endotactic stochastic mass-action system to be transient in a closed communicating class.*

Corollary 3.12. *Every second order* **1***-endotactic stochastic mass-action system is positive recurrent in a closed communicating class* Γ *if and only if there exists a stationary distribution in* Γ*.*

Proof. The forward implication ("only if" part) is obvious and the backward implication ("if" part) follows immediately from [45, Theorem 3.5.3] and Corollary 3.3. □

Remark 3.13. In [1], a 7th order null recurrent strongly endotactic stochastic mass-action system was constructed. Note that every stochastic mass-action system has a *stationary measure* in every closed communicating class [54]. To the best knowledge of the author, it seems unknown *what is the minimal order for an endotactic stochastic mass-action system to admit no stationary distribution (i.e., no finite stationary measure)* in an infinite closed communicating class.

The linear function no longer is a generic Lyapunov function for non-explosivity of third order weakly reversible stochastic mass-action systems.

Example 3.14. Consider the following stochastic mass-action system of species S_1 and S_2

$$\mathcal{R}\colon S_1 + S_2 \underset{\kappa_3}{\overset{\kappa_4}{\rightleftharpoons}} 2S_1 + S_2 \underset{\kappa_1}{\overset{\kappa_2}{\rightleftharpoons}} 2S_1$$

Note that $\mathbb{N}\times\mathbb{N}_0\setminus\{(1,0)\}$ is the unique non-singleton communicating class in $\mathbb{N}_0^2$, and is closed. Since $\mathcal{R}$ is weakly reversible and *deficiency zero*[1], it is complex-balanced [32, Theorem 4A], and hence is non-explosive by [4, Theorem 3]. Let $\mathcal{L}_{\mathcal{R}}$ be the infinitesimal generator associated with $\mathcal{R}$. For every $u \in \mathbb{R}^2_{++}$, let $V_u(x) = x \cdot u^T$. Then for all large $x_1 \in \mathbb{N} \setminus \{1\}$,

$$\mathcal{L}_{\mathcal{R}} V_u(x_1, 0) = \kappa_1 x_1(x_1 - 1)u_2$$

Obviously, V_u is not a Lyapunov function for non-explosivity of $\mathcal{R}$ for any $u \in \mathbb{R}^2_+$ (note that V_u is not *norm-like* for any $u \in \mathbb{R}^2_+ \setminus \mathbb{R}^2_{++}$). Essentially, this is because the reaction $2S_1 \xrightarrow{\kappa_1} 2S_1 + S_2$ of a strictly *super-linear* propensity function increases (the molecule counts of S_2) along any positive direction.

Below we provide an example in action how to apply Corollary 3.12 to prove positive recurrence.

Example 3.15. Consider the following reaction network taken from [30, Example 1]

$$\mathcal{R}\colon 2S_3 \underset{\kappa_2}{\overset{\kappa_1}{\rightleftharpoons}} 0 \xrightarrow{\kappa_3} S_1 \xrightarrow{\kappa_4} S_2 \quad 2S_1 \xrightarrow{\kappa_5} S_1 + S_2 \quad 2S_2 \xrightarrow{\kappa_6} S_2 + S_3$$

This second order stochastic mass-action system neither is $(-1,-1,0)$-endotactic nor *there exists a directed path in the reaction graph from* $2S_1$ *or* $2S_2$ *to either* 0 *or* S_i *for any* $i = 1, 2, 3$ (condition 2 of [6, Theorem 2]). Hence sufficient conditions for positive recurrence in terms of the reaction graph [6, 4, 2, 3] seem arguably not applicable to $\mathcal{R}$, at least in a straightforward manner. However, it is readily verified that $\mathcal{R}_{(1,1,1)} = \{2S_3 \underset{\kappa_2}{\overset{\kappa_1}{\rightleftharpoons}} 0 \xrightarrow{\kappa_3} S_1\}$, and hence $\mathcal{R}$ is $(1,1,1)$-endotactic. Moreover, the CTMC associated with $\mathcal{R}$ is irreducible on $\mathbb{N}_0^3$ with a stationary distribution (see [30] for the closed form of the stationary distribution). By Corollary 3.12, $\mathcal{R}$ is positive recurrent.

[1] *Deficiency* is the number of complexes minus the sum of the dimension of $\mathcal{R}$ and the number of non-singleton strongly connected components of $\mathcal{R}$. For this example, it equals $3 - 2 - 1 = 0$.

Despite endotacticity may be non-trivial to verify in general, it nonetheless is rather straightforward to verify **1**-endotacticity by definition as demonstrated by the above examples, regardless of the dimension or order of the reaction network. We will see in the next section how Theorem 3.1 and Corollary 3.3 apply to various models stemming from real-world applications.

4. Applications

Open reaction networks in the sense of Feinberg [16] refer to those that contain "pseudo-reactions" (i.e., reactions where either the reactant or the product is the zero complex), to account for the case where mass exchange with the ambient is possible. Pseudo-reactions of a zero reactant account for reactions where certain species in the reactor vessel is produced in the feed stream at constant molar concentrations. Similarly, pseudo-reactions of a zero product model removal of the species in the reactant from the reactor vessel by the effluent stream. *Closed reaction systems*, as opposed to *open reaction systems*, assume conservation of mass, in terms of the existence of a conservation law for the reaction network [17, 44]. Since closed reaction systems when modelled stochastically are CTMCs on *finite* state spaces owing to the existence of a conservation law, whose non-explosivity trivially holds. To avoid this triviality while demonstrate the applicability of the results established in the previous section, we will apply our results to various biological models which can be represented as *open stochastic reaction systems*. Throughout this section, to save the effort in notation, *the same symbol $\mathcal{R}$ is slightly abused to only refer to that specific reaction network within each subsection.*

Below models with a star in the superscript were originally proposed as *deterministic* reaction systems in the references, where reactions are assumed to occur in an isothermal homogeneous continuous flow stirred tank reactor (CFSTR), and the biological appropriateness might have been justified therein. It is noteworthy that in contrast, we simply take the reaction mechanism as a reaction network and derive a stochastic analogue, by taking into account the randomness induced by possible *low* abundance of species, *without justifying its biological appropriateness*. In spite of this, empirical evidence does support that certain stochasticity may exist in a few models, e.g., the Lotka-Volterra model [36], the Brusselator model [10], the glycolysis model [33], the enzymatic model [26], and metabolic pathways [38].

It is plausible that one motivation of mathematical biology is to unveil the common mathematical law of similar biological models. Since it is arguably closer to the truth (than the negation of the following statement) that many phenomena in nature share a same generic principle or law, similar or even identical chemical mechanisms may exist in more than one forms in nature. For this consideration, despite the assumption of low abundance justifying the stochastic models seems contradictive to the assumption of high copy numbers of species which underlies deterministic reaction systems, the discussion about the stochastic models of the chemical mechanisms for deterministic reaction systems in the references may not be entirely meaningless, even from the perspective of biology.

Stochastic SIR model. Consider the following susceptible-infected-recovered (SIR) epidemic model [24] in terms of the following reaction network:

$$\mathcal{R}\colon R \xrightarrow{\kappa_{rs}} S \underset{\beta_s}{\overset{\gamma_s}{\rightleftharpoons}} 0 \underset{\gamma_i}{\overset{\beta_i}{\rightleftharpoons}} I \xrightarrow{\kappa_{ir}} R \xrightarrow{\gamma_r} 0 \quad S + I \xrightarrow{\kappa_{si}} 2I$$

where respectively, species S stands for the susceptible, I the infected, and R the recovered compartment of individuals; β_s and β_i are the birth rates of susceptibles and infected individuals; γ_s, γ_i, and γ_r are the death rates of susceptibles, infected individuals as well as recovered individuals; κ_{si} is the infection rate, κ_{rs} is the recovery rate, and κ_{rs} is rate that a recovered individual loses immunity and immediately becomes susceptible again. It is easy

to verify that this reaction network is $(1,1,1)$-endotactic with

$$\mathcal{R}_{(1,1,1),+} = \{S \xleftarrow{\kappa_s} 0 \xrightarrow{\kappa_i} I\},$$

since total molecule counts of species will *not* increase through any of the remaining reactions. Assume low population size. Then the SIR contact process incorporating stochasticity induced by low population size can be modelled as a stochastic mass-action system. By Corollary 3.3, this stochastic SIR model is non-explosive regardless of the choice of the positive rate constants.

Stochastic circadian clock. Consider the circadian clock network of 9 species and 16 reactions [51, 24]:

$$\mathcal{R}\colon \begin{cases} 0 \xleftarrow{\delta_{M_A}} M_A \xleftarrow{\alpha_A} D_A + A \underset{\theta_A}{\overset{\gamma_A}{\rightleftharpoons}} D'_A + A \xrightarrow{\alpha_{A'}} M_A \xrightarrow{\beta_A} A \xrightarrow{\delta_A} 0 \\ 0 \xleftarrow{\delta_{M_R}} M_R \xleftarrow{\alpha_R} D_R + A \underset{\theta_R}{\overset{\gamma_R}{\rightleftharpoons}} D'_R + A \xrightarrow{\alpha_{R'}} M_R \xrightarrow{\beta_R} R \xrightarrow{\delta_R} 0 \\ A + R \xrightarrow{\gamma_C} C \xrightarrow{\delta_A} R \end{cases}$$

where A is the activator protein, R the repressor protein, and C is the protein formed by binding A and R; D_A and D'_A are the activator promoters with and without A bound to them, respectively; D_R and D'_R are the activator promoters with and without R bound to them, respectively; M_A and M_R are mRNA of A and R, respectively.

It is straightforward to verify that $\mathcal{R}$ is **1**-endotactic as there are no reactions increasing total molecule counts of species. By Corollary 3.3, the circadian clock network as a second order stochastic mass-action system is non-explosive.

Brusselator reaction network*. Let us revisit Brusselator reaction network:

$$\mathcal{R}\colon 0 \rightleftharpoons S_1 \longrightarrow S_2 \quad 2S_1 + S_2 \longrightarrow 3S_1$$

By Example 2.2, despite $\mathcal{R}$ is not endotactic, it is $(1,1)$-endotactic. By Corollary 3.3, the stochastic Brusselator mass-action system is non-explosive.

Generalized Togashi-Kaneko model. Consider the generalized Togashi-Kaneko (TK) model [11]:

$$\mathcal{R}\colon S_i + S_j \xrightarrow{\kappa_{ij}} 2S_j \quad S_i \underset{\kappa_{-i}}{\overset{\kappa_i}{\rightleftharpoons}} 0, \quad i,j \in [d],\ i \neq j$$

where the reaction rate constants are *non-negative*, and reactions are removed whenever the corresponding reaction rate constants are set zero. It recently has been discovered that this model undergoes *discreteness-induced transitions* [11]. Observe that $\mathcal{R}$ is of second order, and it is straightforward to verify that $\mathcal{R}$ is **1**-endotactic for the same sake as in the stochastic SIR model. By Corollary 3.3, this generalized TK model as a stochastic mass-action system is non-explosive.

Enzyme reactions in a semi-open reactor*. Consider an open enzymatic reaction network [17, Section 4.2.2]:

$$\mathcal{R}\colon \begin{cases} S_1 + S_3 \underset{\kappa_2}{\overset{\kappa_1}{\rightleftharpoons}} S_4 \quad S_2 + S_3 \underset{\kappa_4}{\overset{\kappa_3}{\rightleftharpoons}} S_5 \\ S_2 + S_4 \underset{\kappa_6}{\overset{\kappa_5}{\rightleftharpoons}} S_6 \underset{\kappa_8}{\overset{\kappa_7}{\rightleftharpoons}} S_1 + S_5 \\ S_6 \xrightarrow{\kappa_9} S_3 + S_7 \\ S_1 \underset{\kappa_{11}}{\overset{\kappa_{10}}{\rightleftharpoons}} 0 \underset{\kappa_{13}}{\overset{\kappa_{12}}{\rightleftharpoons}} S_2 \quad S_7 \xrightarrow{\kappa_{14}} 0 \end{cases}$$

where S_1 and S_2 are substrates, and S_3 stands for an enzyme. In the first step (reactions in the first line), both substrates S_1 and S_2 bind to the enzyme S_3 in respective binding sites to form two intermediate species S_4 and S_5. In the second step (reactions in the second line) S_1 binds reversibly to S_5, and similarly, S_2 binds reversibly to S_4. Once both substrates

are bound to the enzyme in the form of the new intermediate species S_6, the product S_7 is formed (the reaction in the third line). The reactions in the last line refer to the influx of the substrates as well as the efflux of the substrates and the product.

It is easy to verify that $\mathcal{R}$ is **1**-endotactic, and applying Corollary 3.3 yields the stochastic mass-action system modelling this enzymatic reaction network is non-explosive.

An open Peroxidase-oxidase (PO) reaction system*. Consider the following peroxidase-oxidase (PO) reaction in an open system whose mechanism is given by the following termolecular reaction network [46]:

$$\mathcal{R}\colon \begin{cases} S_1+S_2+S_3 \xrightarrow{\kappa_1} 2S_3 \xrightarrow{\kappa_2} 2S_4 \quad S_1+S_2+S_4 \xrightarrow{\kappa_3} 2S_3 \\ 0 \xrightarrow{\kappa_4} S_3 \xrightarrow{\kappa_5} S_5 \quad S_4 \xrightarrow{\kappa_6} S_6 \quad S_1 \underset{\kappa_8}{\overset{\kappa_7}{\rightleftharpoons}} S_7 \quad S_8 \xrightarrow{\kappa_9} S_2 \end{cases}$$

It is readily verified that $\mathcal{R}$ is **1**-endotactic with

$$\mathcal{R}_{\mathbf{1}} = \{0 \xrightarrow{\kappa_4} S_3 \quad S_1+S_2+S_3 \xrightarrow{\kappa_1} 2S_3 \xleftarrow{\kappa_3} S_1+S_2+S_4\}$$

By Corollary 3.3, the stochastic mass-action peroxidase-oxidase (PO) reaction system is non-explosive.

Autocatalytic bromate-bromide-cerium(III) reaction*. It is well-known that oscillations of concentrations of species were observed in Belousov-Zhabotinskii reaction [20]. Consider the autocatalytic bromate-bromide-cerium (III) reaction process whose mechanism as a simplification of the full mechanism given in [21] can be formulated as the following *closed* reaction network:

$$\mathcal{R}\colon \begin{cases} S_1+S_2+2S_3 \xrightarrow{\kappa_1} S_4+S_5 \quad S_2+S_3+S_4 \xrightarrow{\kappa_2} 2S_5 \quad S_1+S_3+S_4 \xrightarrow{\kappa_3} 2S_6+S_7 \\ S_3+S_6+S_8 \xrightarrow{\kappa_4} S_4+S_9 \quad 2S_4 \xrightarrow{\kappa_5} S_1+S_3+S_5 \end{cases}$$

where $S_1 = \mathrm{BrO}_3^-$, $S_2 = \mathrm{Br}^-$, $S_3 = \mathrm{H}^+$, $S_4 = \mathrm{HBrO}_2$, $S_5 = \mathrm{HOBr}$, $S_6 = \mathrm{BrO}_2^-$, $S_7 = \mathrm{H}_2\mathrm{O}$, $S_8 = \mathrm{Ce}^{3+}$, and $S_9 = \mathrm{Ce}^{4+}$. It is easy to verify that $(2,2,1,3,3,2,2,1,1)$ is one conservation law.

To explain the oscillations of molar concentrations of species in experiment, a reduced mechanism leading to an *open* deterministic chemical system was proposed in [22], where molecule counts of all other species but S_2, S_4 and S_9 are regarded as constants. Hence the reduced open reaction network is given as follows:

$$\widetilde{\mathcal{R}}\colon S_2 \xrightarrow{\widetilde{\kappa}_1} S_4 \xrightarrow{\widetilde{\kappa}_3} 0 \xleftarrow{\widetilde{\kappa}_5} 2S_4 \quad S_2+S_4 \xrightarrow{\widetilde{\kappa}_2} 0 \xrightarrow{\widetilde{\kappa}_4} S_4+S_9$$

where the reaction rate constants with a tilde symbol are the new constants based on the assumption for the reduced reaction network.

It is easy to observe that $\widetilde{\mathcal{R}}$ is **1**-endotactic. Applying Corollary 3.3, this stochastic open mass-action reaction system $\widetilde{\mathcal{R}}$ is non-explosive.

Volterra model. Consider the following Volterra model proposed in [24] as a *stochastic* analogue of a generalization of the classical deterministic competition system [52] among d ecological species:

$$\mathcal{R}\colon 0 \underset{\delta_i}{\overset{\alpha_i}{\rightleftharpoons}} S_i \xrightarrow{\beta_i} 2S_i \quad S_i+S_j \xrightarrow{\gamma_{ij}} S_i, \quad i,j=1,\dots,d,$$

where for species S_i, α_i is the immigration rate, β_i the reproduction rate, and δ_i the death/emigration rate; γ_{ij} is the competition rate between species S_i and S_j due to overpopulation. It is readily verified that $\mathcal{R}$ is **1**-endotactic. By Corollary 3.3, this stochastic Volterra mass-action system is non-explosive regardless of the choice of the rate constants.

Below we provide models where Corollary 3.3 fails to apply. Nevertheless, one can still apply Theorem 3.1 to conclude non-explosivity.

Lotka model*. Consider the following classical Lotka model [39]:

$$\mathcal{R}\colon S_1 \xrightarrow{\kappa_1} 2S_1 \quad S_1 + S_2 \xrightarrow{\kappa_2} 2S_2 \quad S_2 \xrightarrow{\kappa_3} 0$$

The deterministic mass-action system associated with this reaction network admits periodic solutions [16]. To see if the associated stochastic mass-action system is non-explosive, first note that the reaction network is *not* $(1,1)$-endotactic and hence Corollary 3.3 does not apply. Nonetheless, since $\mathcal{R}_{\mathbf{1},+} = \{S_1 \xrightarrow{\kappa_1} 2S_1\}$, the sub-nonlinearity condition (3.1) is fulfilled. It follows from Theorem 3.1 that this stochastic mass-action system is non-explosive.

Glycolysis system*. Consider the Glycolysis chemical system [28, 18]

$$\mathcal{R}\colon 0 \xrightarrow{\kappa_1} S_1 \quad S_1 + S_2 \xrightarrow{\kappa_2} S_3 + S_4 \quad S_3 \xrightarrow{\kappa_3} S_2 \underset{\kappa_5}{\overset{\kappa_4}{\rightleftharpoons}} S_4 + S_5 \quad S_4 + S_6 \xrightarrow{\kappa_6} S_7 \xrightarrow{\kappa_7} S_6$$

With no effort, one can show that $\mathcal{R}$ fulfills (3.1) with

$$\mathcal{R}_{\mathbf{1},+} = \{0 \xrightarrow{\kappa_1} S_1 \quad S_2 \xrightarrow{\kappa_4} S_4 + S_5\}$$

By Theorem 3.1, this stochastic mass-action system is non-explosive.

Feedback loop network. Consider the feedback loop network [24]:

$$\mathcal{R}\colon S_3 \underset{\kappa_{23}}{\overset{\kappa_{32}}{\rightleftharpoons}} 2S_2 \quad S_i \xrightarrow{\gamma_i} 0 \xrightarrow{\eta(S_3)} S_1 \xrightarrow{\kappa_2} S_1 + S_2, \quad i = 1,2,3,$$

where S_1 is mRNA, S_2 is the corresponding protein, and S_3 is the protein dimer that regulates the gene expression via a *sub-linear* positive function $\eta\colon \mathbb{R}_+ \to \mathbb{R}_{++}$ in the sense that

$$\kappa_3 := \sup_{a \in \mathbb{R}_{++}} \frac{\eta(a)}{a} < \infty$$

This SRS is generically of *non*-mass-action kinetics (except when η is a constant function), which renders Corollary 3.3 not applicable. Nevertheless, it is readily proved that $\mathcal{R}$ is $(1,1,1)$-endotactic with

$$\mathcal{R}_{(1,1,1),+} = \{S_3 \xrightarrow{\kappa_{32}} 2S_2, \quad 0 \xrightarrow{\eta(S_3)} S_1 \xrightarrow{\kappa_2} S_1 + S_2\}$$

and

$$\sup_{y \to y' \in \mathcal{R}_{(1,1,1),+}} \sup_{x \in \mathbb{N}^d \setminus \{0\}} \frac{\lambda_{y \to y'}(x)}{\|x\|_1} = \max\{\kappa_{32}, \kappa_2, \kappa_3\} < \infty$$

By Theorem 3.1, this SRS is non-explosive.

Synthetic tetracycline-inducible regulatory network. Let us now continue to consider a more involved regulatory network $\mathcal{R}$–a synthetic tetracycline-inducible regulatory network [50] given in Figure 4. It is a termolecular reaction network consisting of 63 species and 93 distinct chemical reactions. In total, there are 27 bimolecular reactions, 9 termolecular reactions, and 57 unimolecular reactions. It is readily observed that all bimolecular and termolecular reactions are *binding* reactions, and thus are decreasing along $\mathbf{1}$-direction. This implies that $\mathcal{R}_{\mathbf{1},+}$ consists of first order reactions only, and hence $\mathcal{R}$ is $\mathbf{1}$-endotactic.

In [50], not all reactions of the reaction network is endowed with stochastic mass-action kinetics, as some with § in the reference column of the table are adjusted for fast reacting intermediates. Even if it might be possible to transform all reactions of non-mass-action kinetics into multi-step reactions of mass-action kinetics, we here prefer to save the effort while do a bit cheating by taking stochastic mass-action kinetics for the entire reaction network. Since $\mathcal{R}$ is of third order, Corollary 3.3 fails. While from Theorem 3.1, we know that the stochastic mass-action system modelling this reaction network $\mathcal{R}$ is non-explosive since condition (3.1) is fulfilled owing to that $\mathcal{R}_{\mathbf{1},+}$ consists of first order reactions.

#	Reaction	k	Ref.	#	Reaction	k	Ref.
1	TcEx --> Tc	3.3E-4	[31]	48	TetOFF2 + OP2 --> TetOFF2:OP2	2.86E+06	[34]
2	2 TetR --> TetR2	1.00E+09	[32]	49	TetOFF2:OP2 --> TetOFF2 + OP2	5.11E-04	[34]
3	TetR2 --> 2 TetR	10	[32]	50	TetR2 + OP2 --> TetR2:OP2	2.86E+06	[34]
4	2 TetOFF --> TetOFF2	1.00E+09	[32]	51	TetR2:OP2 --> TetR2 + OP2	5.11E-04	[34]
5	TetOFF2 --> 2 TetOFF	10	[32]	52	TetOFF2:OP2 + Tc --> Tc:2TetOFF:OP2	2.0E+6	[4]
6	Tc + TetR2 -->Tc:2TetR	2.00E+06	[4]	53	Tc:2TetOFF:OP1 + Tc --> Tc:TetOFF2:OP2	1.0E+15	§
7	Tc + Tc:2TetR --> Tc:TetR2	1.0E+15	§	54	Tc:TetOFF2:OP2 --> Tc:TetOFF2 + OP2	5.8E-3	[35]
8	Tc:TetR2 --> TetR2 + 2Tc	0.001	[4]	55	TetR2:OP2 + Tc --> Tc:2TetR:OP2	2.0E+6	[4]
9	Tc + TetOFF2 --> Tc:2TetOFF	2.00E+06	[4]	56	Tc:2TetR:OP2 + Tc --> Tc:TetR2:OP2	1.0E+15	§
10	Tc + Tc:2TetOFF --> Tc:TetOFF2	1.0E+15	§	57	Tc:TetR2:OP2 --> Tc:TetR2 + OP2	5.8E-3	[35]
11	Tc:TetOFF2 --> TetOFF2 + 2Tc	0.001	[4]	58	RNAp + P2 + OP2 --> RNAp:P2:OP2	8.60E+6	[38]
12	TetOFF -->	0.0012	*	59	RNAp:P2:OP2 --> RNAp + P2 + OP2	0.01	[38]
13	TetOFF2 -->	0.0012	*	60	RNAp:P2:OP2 --> RNAp*:P2:OP2	0.013	[38]
14	TetR -->	0.0012	*	61	RNAp*:P2:OP2 --> RNAp*:DNA21 + P2 + OP2	30	[40]
15	TetR2 -->	0.0012	*	62	RNAp + P2 + OP2:TetR2 --> RNAp:P2:OP2:TetR2	8.60E+6	[38]
16	Tc:TetOFF2 --> Tc	0.0012	*	63	RNAp:P2:OP2:TetR2 --> RNAp + P2 + OP2:TetR2	0.01	[38]
17	Tc:TetR2 --> Tc	0.0012	*	64	RNAp:P2:OP2:TetR2 --> RNAp*:P2:OP2:TetR2	0.013	[38]
18	GFP-LAA -->	2.88E-04	[36]	65	RNAp*:P2:OP2:TetR2 --> RNAp*:DNA21 + P2 + OP2:TetR2	30	[40]
19	Tc -->	2.67E-06	[37]	66	RNAp + P2 + OP2:TetOFF2 --> RNAp:P2:OP2:TetOFF2	8.60E+6	[38]
20	TcEx -->	2.67E-06	[37]	67	RNAp:P2:OP2:TetOFF2 --> RNAp + P2 + OP2:TetOFF2	0.01	[38]
21	TetOFF2 + OP1 --> TetOFF2:OP1	2.86E+06	[34]	68	RNAp:P2:OP2:TetOFF2 --> RNAp*:P2:OP2:TetOFF2	0.13	¶
22	TetOFF2:OP1 --> TetOFF2 + OP1	5.11E-04	[34]	69	RNAp*:P2:OP2:TetOFF2 --> RNAp*:DNA21 + P2 + OP2:TetOFF2	30	[40]
23	TetR2 + OP1 --> TetR2:OP1	2.86E+06	[34]	70	RNAp*:DNA21 --> RNAp + mRNA2	30, 621	[40]
24	TetR2:OP1 --> TetR2 + OP1	5.11E-04	[34]	71	mRNA2 -->	0.002	$
25	TetOFF2:OP1 + Tc --> Tc:2TetOFF:OP1	2.0E+6	[4]	72	mRNA2 + Ribosome --> Rib:mRNA2	1.00E+05	$
26	Tc:2TetOFF:OP1 + Tc --> Tc:TetOFF2:OP1	1.0E+15	§	73	Rib:mRNA2 --> Rib:mRNA21 + mRNA2	100	[41]
27	Tc:TetOFF2:OP1 --> Tc:TetOFF2 + OP1	5.8E-3	[35]	74	Rib:mRNA21 --> Ribosome + TetR	100, 207	[41]
28	TetR2:OP1 + Tc --> Tc:2TetR:OP1	2.0E+6	[4]	75	TetOFF2 + OP3 --> TetOFF2:OP3	2.86E+06	[34]
29	Tc:2TetR:OP1 + Tc --> Tc:TetR2:OP1	1.0E+15	§	76	TetOFF2:OP3 --> TetOFF2 + OP3	5.11E-04	[34]
30	Tc:TetR2:OP1 --> Tc:TetR2 + OP1	5.8E-3	[35]	77	TetR2 + OP3 --> TetR2:OP3	2.86E+06	[34]
31	RNAp + P1 + OP1 --> RNAp:P1:OP1	8.60E+06	[38]	78	TetR2:OP3 --> TetR2 + OP3	5.11E-04	[34]
32	RNAp:P1:OP1 --> RNAp + P1 + OP1	0.01	[38]	79	TetOFF2:OP3 + Tc --> Tc:2TetOFF:OP3	2.0E+6	[4]
33	RNAp:P1:OP1 --> RNAp*:P1:OP1	0.013	[38]	80	Tc:2TetOFF:OP1 + Tc --> Tc:TetOFF2:OP3	1.0E+15	§
34	RNAp*:P1:OP1 --> RNAp*:DNA11 + P1 + OP1	30	[40]	81	Tc:TetOFF2:OP3 --> Tc:TetOFF2 + OP3	5.8E-3	[35]
35	RNAp + P1 + OP1:TetR2 --> RNAp:P1:OP1:TetR2	8.60E+6	[38]	82	TetR2:OP3 + Tc --> Tc:2TetR:OP3	2.0E+6	[4]
36	RNAp:P1:OP1:TetR2 --> RNAp + P1 + OP1:TetR2	0.01	[38]	83	Tc:2TetR:OP3 + Tc --> Tc:TetR2:OP3	1.0E+15	§
37	RNAp:P1:OP1:TetR2 --> RNAp*:P1:OP1:TetR2	0.013	[38]	84	Tc:TetR2:OP3 --> Tc:TetR2 + OP3	5.8E-3	[35]
38	RNAp*:P1:OP1:TetR2 --> RNAp*:DNA11 + P1 + OP1:TetR2	30	[40]	85	RNAp + P3 + OP3 --> RNAp:P3:OP3	8.60E+6	[38]
39	RNAp + P1 + OP1:TetOFF2 --> RNAp:P1:OP1:TetOFF2	8.60E+6	[38]	86	RNAp:P3:OP3 --> RNAp + P3 + OP3	0.01	[38]
40	RNAp:P1:OP1:TetOFF2 --> RNAp + P1 + OP1:TetOFF2	0.01	[38]	87	RNAp:P3:OP3 --> RNAp*:P3:OP3	0.013	[38]
41	RNAp:P1:OP1:TetOFF2 --> RNAp*:P1:OP1:TetOFF2	0.13	¶	88	RNAp*:P3:OP3 --> RNAp*:DNA31 + P3 + OP3	30	[40]
42	RNAp*:P1:OP1:TetOFF2 --> RNAp*:DNA11 + P1 + OP1:TetOFF2	30	[40]	89	RNAp*:DNA31 --> RNAp + mRNA3	30, 723	[40]
43	RNAp*:DNA11 --> RNAp + mRNA1	30, 660	[40]	90	mRNA3 -->	0.002	$
44	mRNA1 -->	0.002	$	91	mRNA3 + Ribosome --> Rib:mRNA3	1.00E+05	$
45	mRNA1 + Ribosome --> Rib:mRNA1	1.00E+05	$	92	Rib:mRNA3 --> Rib:mRNA31 + mRNA3	100	[41]
46	Rib:mRNA1 --> Rib:mRNA11 + mRNA1	100	[41]	93	Rib:mRNA31 --> Ribosome + GFP-LAA	100, 248	[41]
47	Rib:mRNA11 --> Ribosome + TetOFF	100, 220	[41]				

FIGURE 1. Adapted from [50, Table 1]. The figure represents a synthetic tetracycline-inducible regulatory network of 93 reactions. Bimolecular reactions are colored yellow and termolecular reactions are colored green. To read this table with more details, the reader is referred to [50].

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Appendix A. Drift criteria for non-explosivity

Proposition A.1. *[42, Theorem 2.1] Let X_t be a CTMC on an infinite state space $\mathcal{X} \subseteq \mathbb{Z}^d$. Assume there exist constants $C_1, C_2 \geq 0$, and a non-negative function $V\colon \mathcal{X} \to \mathbb{R}$ such that $\lim_{\|x\|_1 \to \infty} V(x) = +\infty$ and*

$$\mathcal{L}V(x) \leq C_1 V(x) + C_2, \quad \forall x \in \mathcal{X}$$

Then X_t is non-explosive on $\mathcal{X}$.

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